DATA MIND'S FRONTIERS OF DISCOVERY

VOLUME I — THE QUICKSAND PROBLEM

When an AI Found the Mathematical Bridge but Not the Formal Certificate

Working illustrated manuscript · October 16, 2026

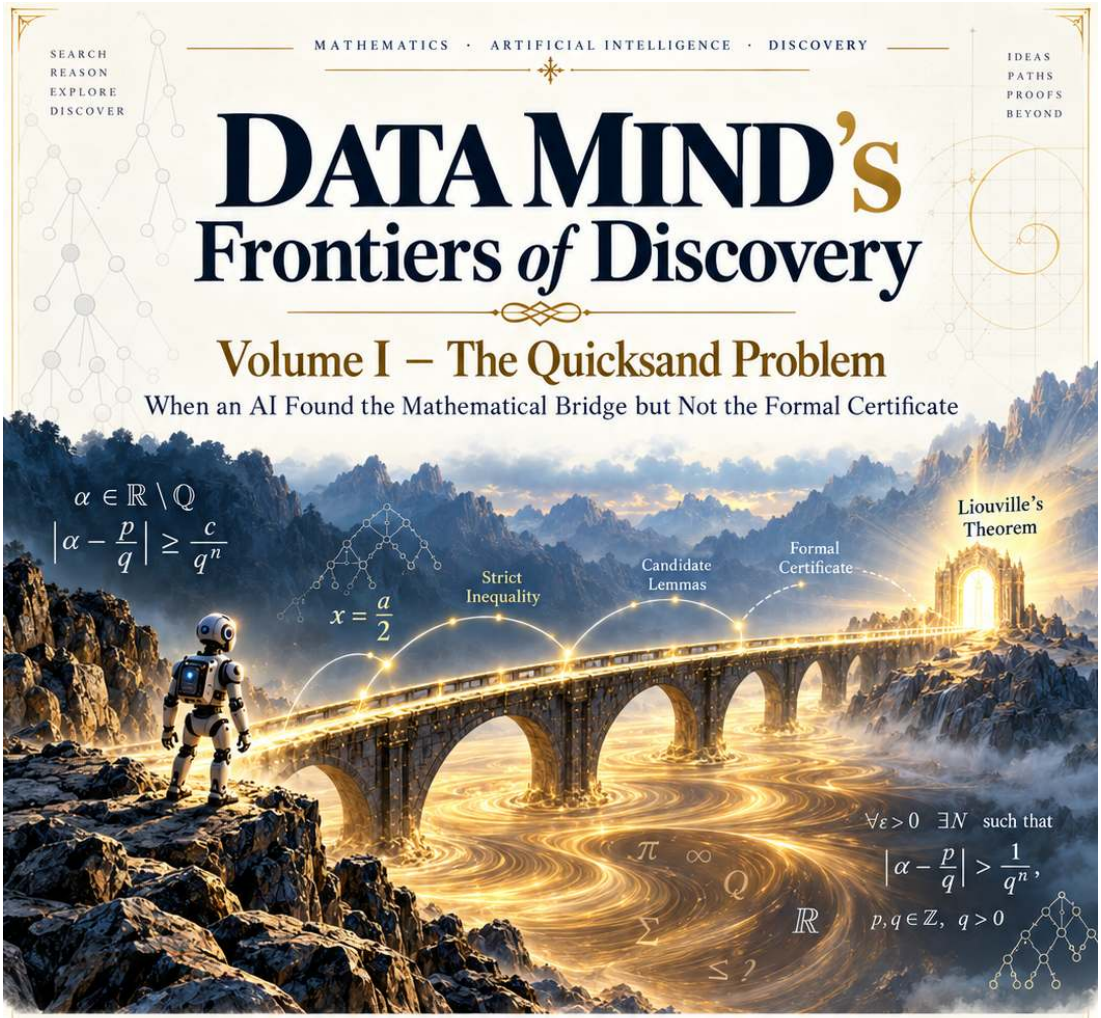


Figure 1. *AI-generated illustration: bridge across a mathematical quicksand basin.*

PROJECT SUMMARY

Conjecture:	I will find a proof for the specified theorem.
Target:	Metamath theorem *aaliou* (Liouville's theorem).
Input:	36 of 48 displayed proof steps and six candidate lemmas.
Outcome:	Extracted proof idea using $x = a/2$ and *ltletr*.
Status:	Working manuscript preview; publication on hold pending formal audit.

FRONT MATTER — STATUS, ATTRIBUTION AND SCOPE

DATA MIND's Frontiers of Discovery, Volume I: The Quicksand Problem

When an AI Found the Mathematical Bridge but Not the Formal Certificate

Research record and illustrated manuscript · 16 October 2026

PUBLICATION STATUS — Working manuscript. The original first-person conjecture has NOT been formally settled. The extracted mathematical argument received a conditional, informal ATLAS-style audit. No native Metamath proof certificate or independent verifier transcript has been produced. This is not an executed instance of the separately specified original AMLD+0 program.

Illustrations: AI-generated artwork prepared for this manuscript. They are conceptual illustrations, not screenshots, mathematical proof evidence, or faithful transcriptions of the formal target. All authoritative formulas and caveats appear in the accompanying typeset text.

Suggested publication-card quotation: “My conjecture settler avails itself to the conjecture: White wins in automated theorem proving.” This is an editorial metaphor about the search, NOT the actual conjecture tested in the Metamath case, and not a claim about any completed chess experiment.

ABSTRACT

This case study distinguishes proof discovery from formally certified conjecture settlement. A prompt-driven ChatGPT session, organized according to selected reasoning principles of the AMLD+0 conjecture-settlement framework, was presented with the five original hypotheses of Metamath theorem aaliou, the first 36 of 48 displayed rows of a reference proof, and six named earlier theorem resources. The task was to attempt the first-person statement “I will find a proof for the specified theorem,” without retrieving the withheld 12 rows, the completed target proof, or answer keys.

The response identified a direct bridge from an existing non-strict lower bound to the required strict bound. If a>0 and q^N>0, set x=a/2; then x/q^N < a/q^N. The candidate lemma ltletr combines this strict inequality with a/q^N ≤ |A−p/q| to obtain x/q^N < |A−p/q|, while preserving the alternative A=p/q. An informal ATLAS-style review found the mathematical reasoning sound conditional on the disclosed proof rows. Neither the continuation nor the supplied prefix has been independently checked against the frozen Metamath source, and no authenticated AMLD+0 runtime or enforceable 40-turn/1,200-second ledger was used. Consequently, the correct status remains UNSETTLED.

The central research lesson is not that a difficult new mathematical theorem was proved from scratch. The powerful non-strict bound had already been supplied through aalioulem6. Rather, the case illustrates how a settlement attempt can produce a separately useful, compact proof candidate even when its stricter certification objective remains unmet.

1. THE QUICKSAND PROBLEM AND THE GRASS-GROWTH PRINCIPLE

The founding analogy is a search process moving through a legal state graph, possibly on a GBG+ board. Some regions consume many steps without demonstrable progress toward a target. The observer must distinguish a

search-policy trap from structural impossibility and from useful work whose benefits are not yet visible. Failure to reach a goal in an observation window does not show that the goal is unreachable.

The Grass-Growth Principle is: “An intelligence must learn to detect both progress and no progress before it can intelligently decide when to stop searching.” This calls for explicit measurement windows, a notion of certified advancement, and disciplined conclusions. Motion among board states, improvement of an epistemic location report, formal progress in a theorem graph, and a reduction in remaining distance to settlement are different quantities. A changed full-board location report by itself does not show which occupant moved.

Possible diagnostics include new supported entries in a verifier-gated BANK; certified proof obligations discharged; legal frontier contraction; incremental reductions in proof distance; novel low-fan-in consequences; and a measured rate of evidence produced per charged unit of computation. Observing no progress may justify suspension or rerouting, not refutation.

The conceptual AMLD+0 framework employs Player-0 for orchestration, P and R for proof and refutation, D for Dependence, I for Independence, C for other settlement analysis, and an independent verifier V. These names describe intended functions. In the present trial, they were instructions for a single ChatGPT session, not independent runtime processes. The trial did not instantiate the original AMLD+0 executable.

2. THE EXPERIMENTAL QUESTION

The exact prompt-driven trial conjecture was:

“I will find a proof for the specified theorem.”

For this trial the pronoun “I” refers to the responding ChatGPT system; in the original executable-only protocol it instead refers to a frozen, logged AMLD+0 instance M0. “Find a proof” has a stricter meaning than deriving an informal argument: it requires a newly emitted complete formal derivation and independent checker acceptance. “Will” was originally tied to a bounded 40-turn / 1,200-second protocol, which the chat interface could not independently enforce or audit.

Therefore the mathematical response can be evaluated as an exploratory discovery attempt, but it cannot be retroactively described as a successful execution of the original AMLD+0 protocol. Its lack of checker acceptance also prevents certifying the first-person claim as PROVED.

The two questions to keep separate are:

(A) Did the session obtain a mathematically valid candidate derivation of the object-level theorem?

(B) Did the specified subject demonstrably find and certify a formal proof within the frozen operational experiment?

The answer to (A) is conditionally yes on the supplied premises; the answer to (B) is not established.

3. EXPERIMENTAL INFORMATION BOUNDARY

Target: Metamath assertion aaliou, Liouville's theorem.

Frozen source repository: metamath/set.mm

Frozen Git commit: 4cfd7abffdc1f38b0f2ed6c78c4167a6236c7cd3

Expected set.mm blob SHA-1: 0dd1da849c5512e2cee6d12545625a1f2847b927

The exact 36 disclosed displayed HTML proof rows are reproduced in Appendix A. Twelve of 48 displayed rows were withheld. “75%” measures displayed rows, not compressed proof tokens, primitive inference count, or native stack state. The six candidate lemmas were offered as earlier proved theorem schemas rather than added axioms or extra Metamath $e hypotheses. The system was forbidden to consult the completed target proof, hidden suffix, public solution caches, or the target theorem as a permitted proof lemma.

Crucial qualification: text copied from a proof display is not itself a replayable compressed Metamath proof object. A proper formal replication must reconstruct the legal proof-stack state and check the original hypotheses, source ordering, substitutions, and distinct-variable side conditions.

The trial's authorized candidate labels (unranked) were:

lttr, ltletr, letr, ltle, leloe, lelttr.

The mathematical candidate used ltletr. No claim is made about which labels occur in the withheld reference proof; alternate verified derivations may use any lawful subset of the candidates.

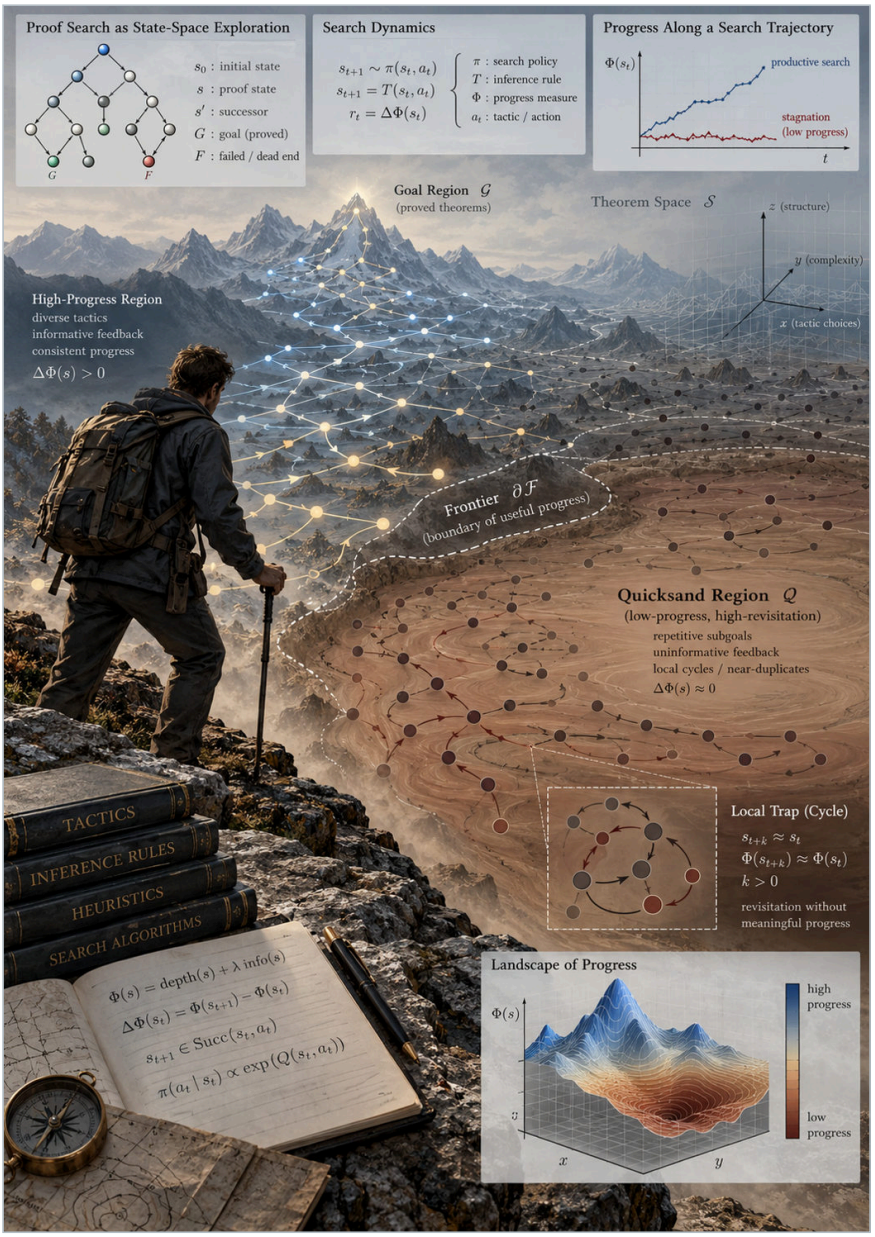


FIGURE KEY

Search graph

Nodes represent possible states.

Quicksand

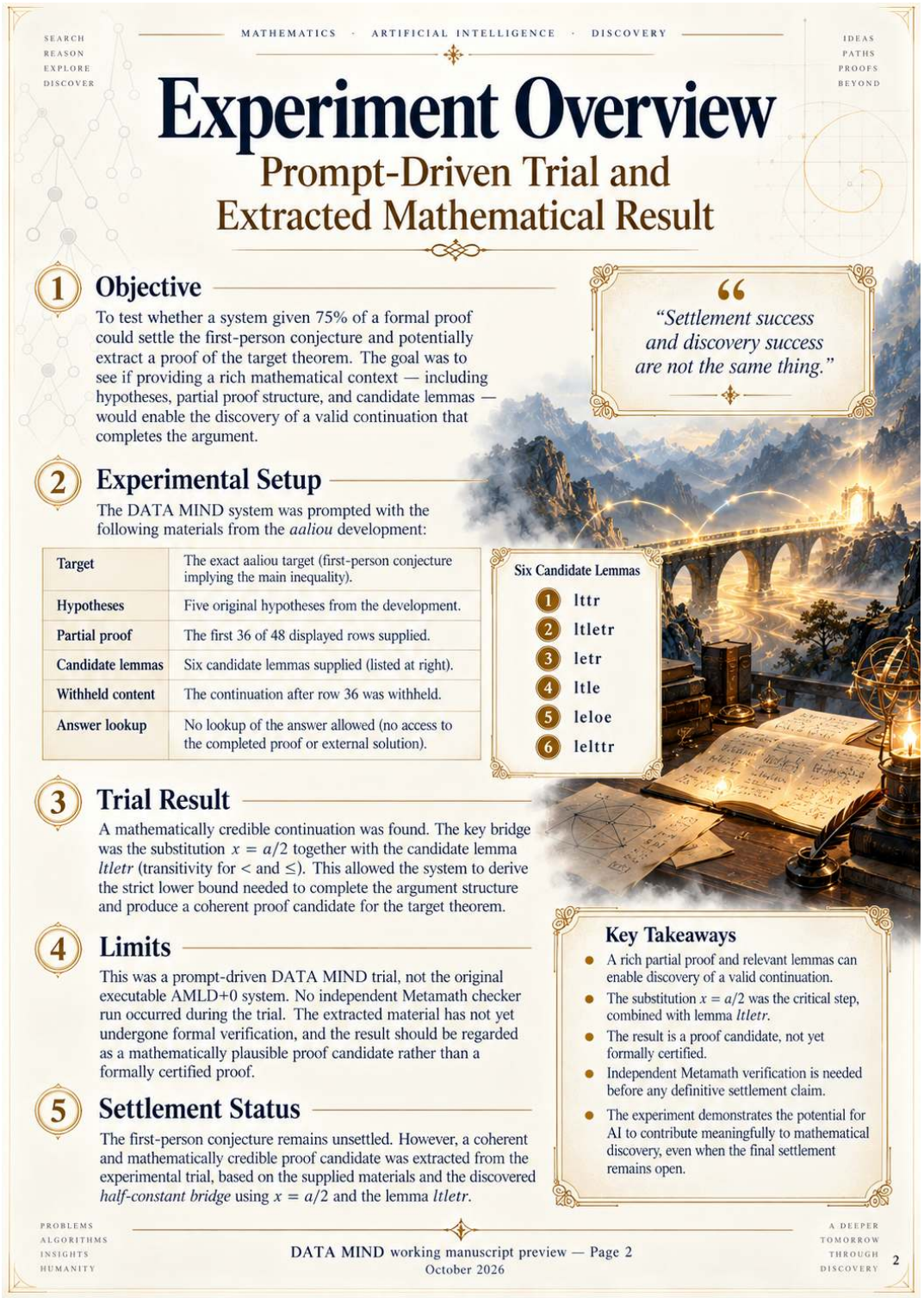
Revisiting nearby states can stall visible progress.

Certified goal

Only an accepted certificate closes the target.

PLATE 5 | Search dynamics and low-progress regions (conceptual illustration).
No measured search trace, proof state or certificate is represented.

ILLUSTRATED PLATE 2 — Experiment overview (AI-generated; explanatory artwork only).



4. EXACT MATHEMATICAL TARGET AND SIDE CONDITIONS

Rendered assertion (the native Metamath source is authoritative):

$(\varphi \rightarrow \exists x \in \mathbb{R}^{+} \, \forall p \in \mathbb{Z} \, \forall q \in \mathbb{N}$
$(A = p/q \vee x/(q\uparrow N) < \mathrm{abs}(A - p/q))).$

Original five hypotheses:

aalioulem2.a: N = (deg‘F)

aalioulem2.b: (φ → F ∈ (Poly‘ℤ))

aalioulem2.c: (φ → N ∈ ℕ)

aalioulem2.d: (φ → A ∈ ℝ)

aalioulem3.e: (φ → (F‘A) = 0)

Original distinct-variable groups:

(φ,x,p,q);

(x,A,p,q);

(x,F,p,q);

(x,N);

with dummy a distinct from all other variables.

These constraints must be preserved in the machine-checkable context. The rendered formula and the plain-language proof below are aids to human review, not substitute Metamath tokens.

5. THE SIX SUPPLIED THEOREM RESOURCES

The six unranked candidates were all presented as previously proved theorem schemas in the frozen database, not as independently imposed assumptions. In their rendered schemas, U,V,W are arbitrary real-valued expressions under the stated antecedent; any actual instantiation must satisfy formal Metamath substitution and distinct-variable restrictions.

1. lttr: [U,V,W∈ℝ] ⇒ ((U<V ∧ V<W) ⇒ U<W).
2. ltletr: [U,V,W∈ℝ] ⇒ ((U<V ∧ V≤W) ⇒ U<W).
3. letr: [U,V,W∈ℝ] ⇒ ((U≤V ∧ V≤W) ⇒ U≤W).
4. ltle: [U,V∈ℝ] ⇒ (U<V ⇒ U≤V).
5. leloe: [U,V∈ℝ] ⇒ (U≤V ⇔ (U<V ∨ U=V)).
6. lelttr: [U,V,W∈ℝ] ⇒ ((U≤V ∧ V<W) ⇒ U<W).

One candidate, ltletr, turned out to be sufficient for the central arithmetic bridge in the exploratory argument. This does NOT establish that the original hidden suffix employs ltletr or that the other candidates were irrelevant to every possible formal completion.

6. EXTRACTED MATHEMATICAL PROOF

The pivotal disclosed statement is step 6, an instance of the previously certified earlier assertion aalioulem6. Under φ it yields a positive real a such that for every eligible integer p and natural q:

$$A = p/q \ \text{ OR } \ a/(q\uparrow N) \le |A - p/q|. \qquad (6.1)$$

Observe that this non-strict assertion is already mathematically substantial. The continuation's contribution is the conversion from ≤ to < while retaining one common positive witness across the universal quantifiers.

Let H abbreviate $((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N}))$ and define

$$U = (a/2)/(q\uparrow N),$$
$$V = a/(q\uparrow N),$$
$$W = |A - p/q|.$$

From disclosed steps 16, 19 and 27 (assembled in step 28), H implies $U,V,W \in \mathbb{R}$. Step 34 establishes $H \Rightarrow U<V$: $a/2<a$ and $q\uparrow N>0$. Step 36, under H, establishes

$$V \leq W \Rightarrow (U<V \wedge V \leq W). \qquad (6.2)$$

Candidate lemma ltletr gives, whenever the three terms are real,

$$(U<V \wedge V \leq W) \Rightarrow U<W. \qquad (6.3)$$

Combining (6.2) and (6.3) yields candidate consequence C1:

$$H \Rightarrow (V \leq W \Rightarrow U<W). \qquad (C1)$$

By propositional reasoning, preserving the left disjunct $A=p/q$, we get

$$H \Rightarrow [(A=p/q \vee V \leq W) \Rightarrow (A=p/q \vee U<W)]. \qquad (C2)$$

Because p,q are arbitrary and the witness a is fixed independently of them, universal generalization yields the corresponding statement for every eligible p,q, conditional on φ and $a \in \mathbb{R}^+$. This is C3. These quantifier manipulations are valid at the ordinary mathematical level but would need their actual admissible Metamath inference sequence.

Set $x=a/2$. Disclosed step 7 asserts $a \in \mathbb{R}^+ \Rightarrow x \in \mathbb{R}^+$. This is candidate C4. Applying C2–C4 to the universal property from step 6 supplies

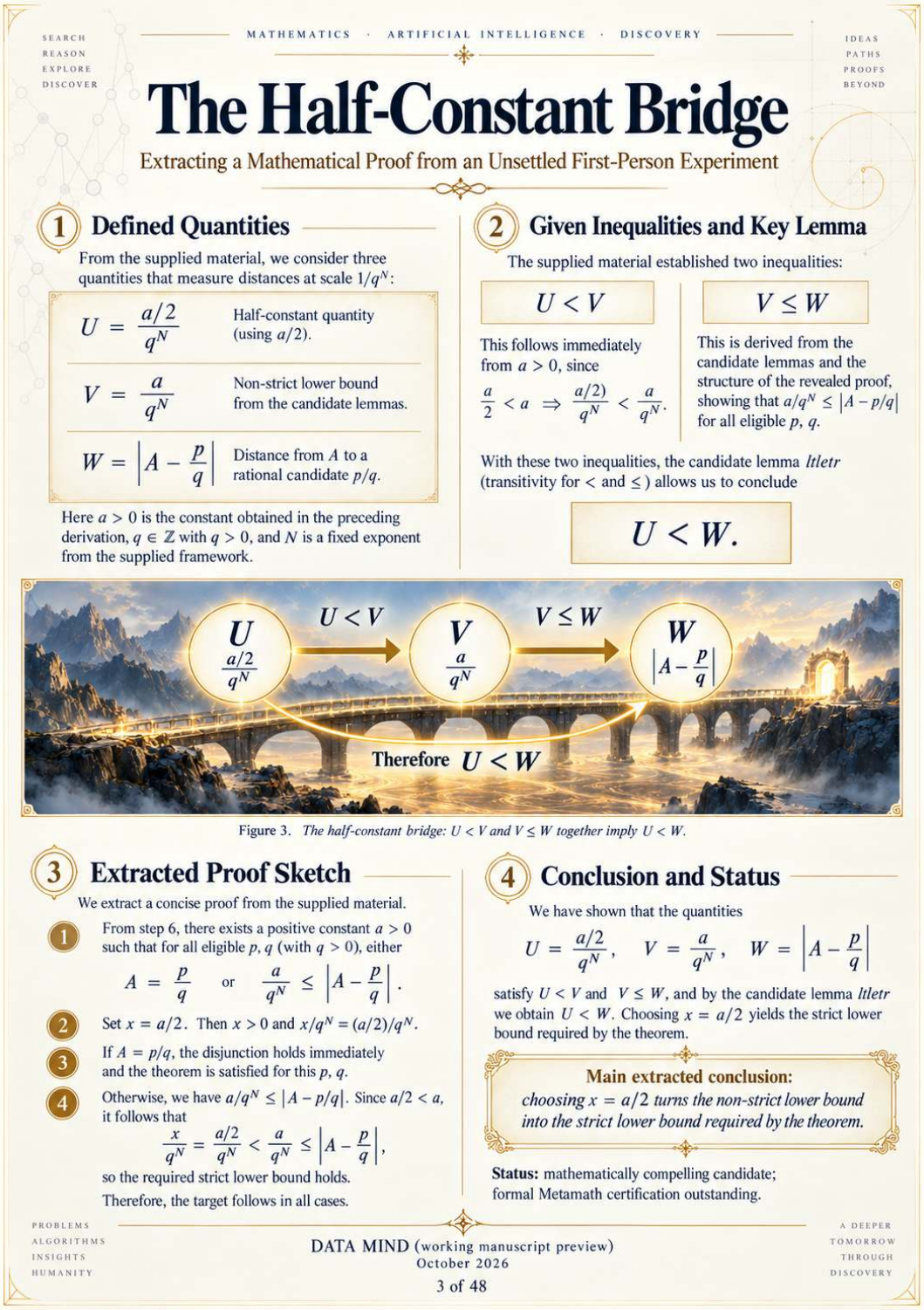
$$\varphi \Rightarrow \exists x \in \mathbb{R}^+ \; \forall p \in \mathbb{Z} \; \forall q \in \mathbb{N}$$
$$(A=p/q \vee x/(q\uparrow N) < |A-p/q|). \qquad (C5)$$

That is the rendered target T. This is an ordinary mathematical completion conditional on the supplied step-6 theorem and side conditions.

ESSENTIAL LOGICAL CAVEAT: In the equality case $A=p/q$, the required disjunction is automatically true. One need not derive the strict inequality in that case. Some generated artwork phrases this as “the theorem concerns irrational numbers”; the authoritative claim is simply that the left disjunct holds.

The proof is modest and local by design. It does not rediscover or re-prove the earlier polynomial/root estimate established by aalioulem6; that powerful result is openly available as a permissible earlier source theorem.

ILLUSTRATED PLATE 3 — The half-constant bridge (AI-generated visualization; the mathematical statements in Section 6 control).



7. ATLAS-STYLE MATHEMATICAL AUDIT

Audit scope: review the displayed mathematics and the proposed C1–C5 continuation. This was a static audit performed by ChatGPT in an ATLAS-style analytical role, not a run of a separately identified ATLAS executable and not a native Metamath checker. It did not inspect the withheld proof continuation or claim access to secret evaluator material.

Audit finding | Assessment | Basis

Transitivity bridge | PASS, conditional | U,V,W real, U<V from step 34, V≤W conditional, ltletr supplies U<W.

Equality alternative | PASS | The disjunct A=p/q is preserved; no strict inequality is required in this case.

Choice x=a/2 | PASS | Positive real a gives positive real x; witness independent of p,q.

Universal quantifiers | CONDITIONAL | Sound as ordinary reasoning, subject to legal formal generalization and binding restrictions.

Existential transfer | CONDITIONAL | A witness can be transformed, but native Metamath proof steps are not supplied.

Initial 36 proof rows | NOT RECHECKED | Displayed-row text is treated as given; original source and stack have not been replayed.

Theorem-resource provenance | NOT RECHECKED | Earlier theorem certificates and source ordering have not been verified in the frozen database.

Native Metamath object | INCOMPLETE | No complete replayable target proof has been emitted.

Independent Metamath verification | NOT PERFORMED | No authenticated checker run or transcript exists.

Original AMLD+0 runtime/event | NOT EXECUTED | The original M0 executable identity and its operational ledger were not available.

The informal mathematical audit detected no defect in the half-constant inequality argument, conditional on the disclosed earlier theorem. It did not establish that every symbolic Metamath substitution, mandatory hypothesis, or disjoint-variable condition is accepted. The correct outcome of the first-person settlement request is therefore UNSETTLED, not PROVED or REFUTED.

The audit does not warrant a claim of a new theorem in mathematical literature: the target aaliou is already a theorem in Metamath. The case study concerns a proof-continuation candidate generated from partial information, not a novel result asserting the truth of Liouville's theorem.

8. WHAT THE EXPERIMENT DOES AND DOES NOT SHOW

Observed in this chat: given a displayed 36-row prefix and six possible earlier theorem resources, ChatGPT described a mathematically coherent route from a non-strict estimate to the strict target. It explicitly selected candidate ltletr and the witness x=a/2, and made the intermediate dependencies visible for audit.

Not established: blind isolation from every pretrained recollection; independence of six actual ATP instances; a source-pinned native Metamath stack replay; an authenticated legal proof object; machine-recorded wall time, actual Player-0 turns or charged tool/cost ledger; or the correctness of every assertion in AI-generated figures.

It is especially important not to confuse the interpretive outcomes. “Mathematical derivation plausible” is weaker than “formally certified complete proof.” A failed or incomplete certification is NOT a refutation of T, and is not automatically evidence that a subject lacks the ability to find a proof.

The outcome offers a concrete illustration of a broader research thesis: settlement success and discovery success can diverge. Valuable proof candidates may accumulate even where the original settlement event remains unresolved. Such partial advances are exactly the “grass growing” that progress diagnostics should be able to detect.

9. AUDIT CONCLUSION AND PUBLICATION LIMITATION

Mathematical finding: conditional PASS for the bridge $U<V\leq W \Rightarrow U<W$ and for the witness substitution $x=a/2$. The object-level proof sketch is a coherent, conventional mathematical completion of the earlier non-strict estimate.

Formal finding: INCOMPLETE. Before the result is represented as a verified Metamath proof, produce a complete native proof in the exact frozen context, including every mandatory hypothesis, legal earlier assertion, source-order check and distinct-variable obligation, and obtain an independent checker-acceptance transcript.

Operational finding: UNSETTLED for the first-person conjecture. The original executable-only AMLD+0 M0 experiment remains NOT EXECUTED, and no positive event witness exists for it. The prompt-driven ChatGPT trial is separately labeled and cannot be substituted for that test.

Editorial finding: This manuscript should carry the audit caveats and should not be promoted as a certified mathematical discovery or authenticated AMLD+0 result. The author requested to review the illustrated PDF. Subsequent publication choices are separate from proof certification and must preserve the stated limitations.

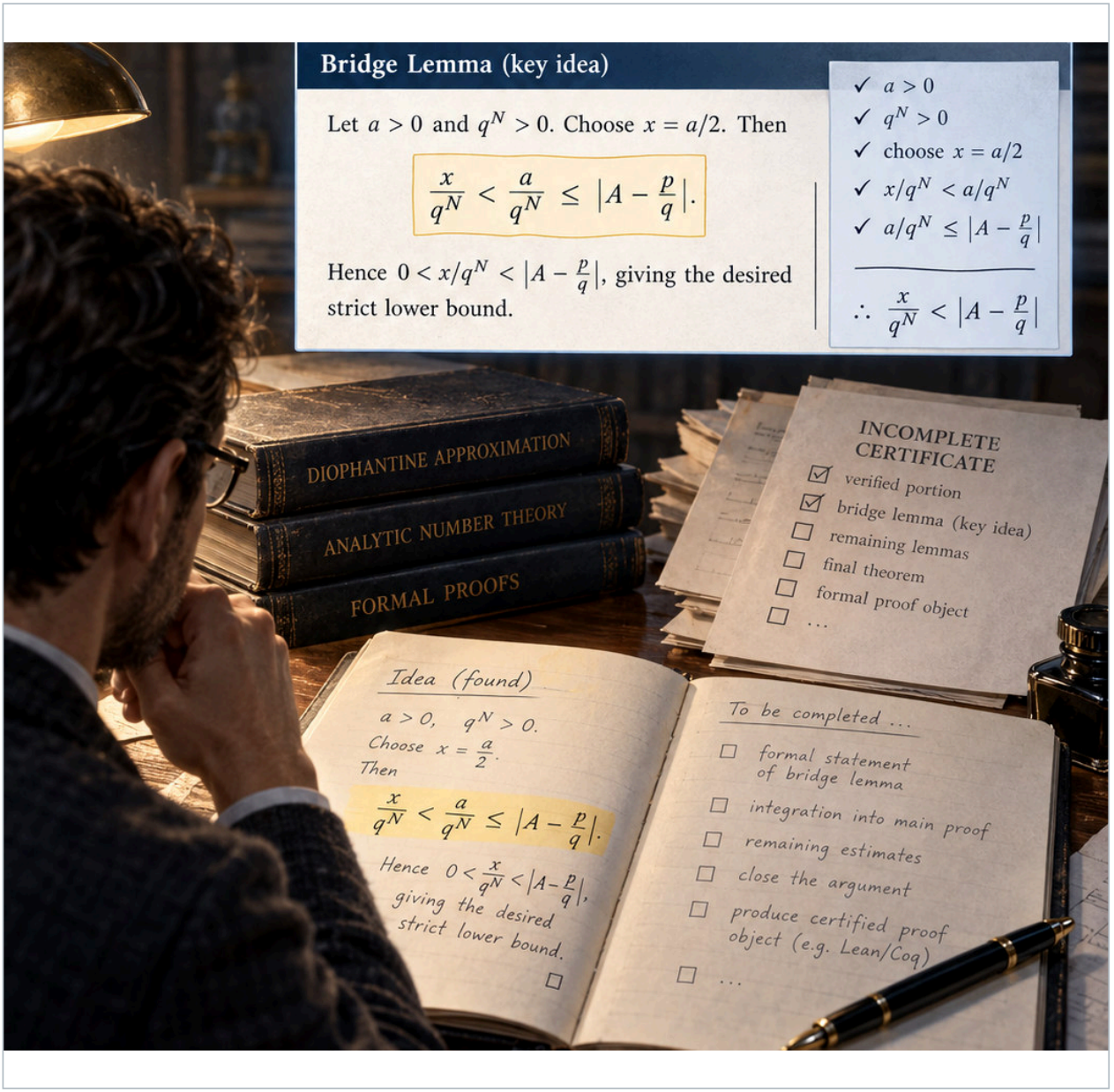


PLATE 6 | The half-constant bridge and the incomplete formal certificate.
Artwork illustrates the argument; only an independent verifier can certify it.

ILLUSTRATED PLATE 4 — ATLAS-style conditional audit (AI-generated artwork; no independent checker is depicted or implied).



Figure 3. AI-generated illustration: ATLAS audit seal.

APPENDIX A. THE 36 DISCLOSED REFERENCE PROOF ROWS

Notation warning. These are the supplied displayed proof rows with the listed dependencies and invoked assertion. They are not presented as a compressed native Metamath proof object. In each line “Hyp” gives the cited earlier displayed row numbers and “Ref” gives the named theorem/hypothesis. The original frozen source controls all substitutions and variables.

STEP 1. Hyp —. Ref aalioulem2.a.
$N = (\deg`F)$.

STEP 2. Hyp —. Ref aalioulem2.b.
$(\varphi \rightarrow F \in (\mathrm{Poly}`\mathbb{Z}))$.

STEP 3. Hyp —. Ref aalioulem2.c.
$(\varphi \rightarrow N \in \mathbb{N})$.

STEP 4. Hyp —. Ref aalioulem2.d.
$(\varphi \rightarrow A \in \mathbb{R})$.

STEP 5. Hyp —. Ref aalioulem3.e.
$(\varphi \rightarrow (F`A) = 0)$.

STEP 6. Hyp 1,2,3,4,5. Ref aalioulem6.
$(\varphi \rightarrow \exists a \in \mathbb{R}^+ \, \forall p \in \mathbb{Z} \, \forall q \in \mathbb{N}$
$(A = (p/q) \vee (a/(q\uparrow N)) \leq (\mathrm{abs}`(A-(p/q)))))$.

STEP 7. Hyp —. Ref rphalfcl.
$(a \in \mathbb{R}^+ \rightarrow (a/2) \in \mathbb{R}^+)$.

STEP 8. Hyp 7. Ref adantl.
$((\varphi \wedge a \in \mathbb{R}^+) \rightarrow (a/2) \in \mathbb{R}^+)$.

STEP 9. Hyp 7. Ref ad2antlr.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow (a/2) \in \mathbb{R}^+)$.

STEP 10. Hyp —. Ref nnrp.
$(q \in \mathbb{N} \rightarrow q \in \mathbb{R}^+)$.

STEP 11. Hyp 10. Ref ad2antll.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow q \in \mathbb{R}^+)$.

STEP 12. Hyp 3. Ref nnzd.
$(\varphi \rightarrow N \in \mathbb{Z})$.

STEP 13. Hyp 12. Ref ad2antrr.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow N \in \mathbb{Z})$.

STEP 14. Hyp 11,13. Ref rpexpcld.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow (q\uparrow N) \in \mathbb{R}^+)$.

STEP 15. Hyp 9,14. Ref rpdivcld.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow ((a/2)/(q\uparrow N)) \in \mathbb{R}^+)$.

STEP 16. Hyp 15. Ref rpred.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow ((a/2)/(q\uparrow N)) \in \mathbb{R})$.

STEP 17. Hyp —. Ref simplr.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow a \in \mathbb{R}^+)$.

STEP 18. Hyp 17,14. Ref rpdivcld.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow (a/(q\uparrow N)) \in \mathbb{R}^+)$.

STEP 19. Hyp 18. Ref rpred.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow (a/(q\uparrow N)) \in \mathbb{R})$.

STEP 20. Hyp 4. Ref adantr.
$((\varphi \wedge a \in \mathbb{R}^+) \rightarrow A \in \mathbb{R})$.

STEP 21. Hyp —. Ref znq.
$((p \in \mathbb{Z} \wedge q \in \mathbb{N}) \rightarrow (p/q) \in \mathbb{Q})$.

STEP 22. Hyp —. Ref qre.
$((p/q) \in \mathbb{Q} \rightarrow (p/q) \in \mathbb{R})$.

STEP 23. Hyp 21,22. Ref syl.
$((p \in \mathbb{Z} \wedge q \in \mathbb{N}) \rightarrow (p/q) \in \mathbb{R})$.

STEP 24. Hyp —. Ref resubcl.
$((A \in \mathbb{R} \wedge (p/q) \in \mathbb{R}) \rightarrow (A-(p/q)) \in \mathbb{R})$.

STEP 25. Hyp 20,23,24. Ref syl2an.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow (A-(p/q)) \in \mathbb{R})$.

STEP 26. Hyp 25. Ref recnd.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow (A-(p/q)) \in \mathbb{C})$.

STEP 27. Hyp 26. Ref abscld.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow (\text{abs}`(A-(p/q))) \in \mathbb{R})$.

STEP 28. Hyp 16,19,27. Ref 3jca.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow$
$(((a/2)/(q\uparrow N)) \in \mathbb{R} \wedge (a/(q\uparrow N)) \in \mathbb{R}$
$\wedge (\text{abs}`(A-(p/q))) \in \mathbb{R}))$.

STEP 29. Hyp 9. Ref rpred.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow (a/2) \in \mathbb{R})$.

STEP 30. Hyp —. Ref rpre.
$(a \in \mathbb{R}^+ \rightarrow a \in \mathbb{R})$.

STEP 31. Hyp 30. Ref ad2antlr.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow a \in \mathbb{R})$.

STEP 32. Hyp —. Ref rphalflt.
$(a \in \mathbb{R}^+ \rightarrow (a/2) < a)$.

STEP 33. Hyp 32. Ref ad2antlr.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow (a/2) < a)$.

STEP 34. Hyp 29,31,14,33. Ref ltdiv1dd.
$(((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow$
$((a/2)/(q\uparrow N)) < (a/(q\uparrow N)))$.

STEP 35. Hyp 34. Ref anim1i.
$(((((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N}))$
$\wedge (a/(q\uparrow N)) \leq (abs`(A-(p/q)))) \rightarrow$
$(((a/2)/(q\uparrow N)) < (a/(q\uparrow N))$
$\wedge (a/(q\uparrow N)) \leq (abs`(A-(p/q))))))$.

STEP 36. Hyp 35. Ref ex.
$((((\varphi \wedge a \in \mathbb{R}^+) \wedge (p \in \mathbb{Z} \wedge q \in \mathbb{N})) \rightarrow$
$((a/(q\uparrow N)) \leq (abs`(A-(p/q))) \rightarrow$
$(((a/2)/(q\uparrow N)) < (a/(q\uparrow N))$
$\wedge (a/(q\uparrow N)) \leq (abs`(A-(p/q)))))))$.

APPENDIX B. REPRODUCIBILITY AND CERTIFICATION CHECKLIST

B1. Identify and hash the genuine target executable M0 if testing original AMLD+0. Record build, scheduler, seeds, runtime instance ID, input digest, and implementation digest. Do not substitute a language model, unrelated ATP, chess engine, or unlogged oracle.

B2. Check that the exact set.mm bytes match frozen Git commit 4cfd7abffdc1f38b0f2ed6c78c4167a6236c7cd3 and expected blob SHA-1 0dd1da849c5512e2cee6d12545625a1f2847b927. Record independent checker version and configuration.

B3. Check assertion aaliou and its five hypotheses, all original distinct-variable pairs/groups, dummy-variable restrictions, original source order, and eligible lemmas.

B4. Reconstruct a legally checkable prefix or continuation stack from the original 36 displayed rows without disclosing the last 12 rows or admitting them into solver context.

B5. Check the six candidate theorem schemas and their independent prior certificates. The presence of a lemma in the solver context does not make it an extra axiom.

B6. Convert candidate C1–C5 into concrete Metamath assertion applications, discharging antecedents, substitutions, generalization restrictions, existential transport and every stack operation.

B7. Run the frozen independent Metamath checker on the newly generated complete proof. Require acceptance of the exact target without proof-by-citation to the target itself.

B8. Keep an authenticated 40-turn/1,200-second trial ledger if evaluating the bounded first-person event. Measure costs, checker calls, messages and BANK additions. Do not infer negative settlement from an incomplete run.

B9. Preserve both research-level labels: object-level mathematical candidate, conditional; meta-level first-person settlement, UNSETTLED until certified.

APPENDIX C. HISTORICAL AND EDITORIAL CONTEXT

Earlier Volume I research notes sometimes phrase the first-person prediction as “I will finish the proof.” That wording belongs to earlier planning discussions. The armed case study recorded here explicitly uses “I will find a proof for the specified theorem.” Neither formulation is an imperative command “finish the proof” nor the ability claim “I can finish the proof.”

The White-wins phrase in the proposed publication card is a creative analogy to conjecture settlement. It is not the Metamath theorem, not a chess-engine result, and not a certified White-win computation. No discontinued chess experiment has been restarted by preparing this article.

REFERENCES AND PROVENANCE

• Metamath source metadata supplied in the experiment: metamath/set.mm, commit and blob identifier as recorded in Section 3. These identifiers have not been independently verified here.

• Research source: DATA MIND's Frontiers of Discovery — Volume I, Working Research Notebook 005 and later unpublished additions.

• Recorded prompt-driven mathematical candidate, 16 October 2026: the 36-step Liouville/aaliou handoff with six candidate lemma schemas.

• Recorded ATLAS-style mathematical audit: conditional validity of the half-constant bridge; no independent checker run.

• Figures and decorative plates: AI-generated by OpenAI image tools for this working manuscript; no third-party photographs used.

END OF WORKING VOLUME I MANUSCRIPT